LaTeX Notes Template

Laurin Seeholzer

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Contents

1 Logic, Sets and Numbers **2**
1.1 Logic 2
1.2 Sets 3
1.3 Numbers 3
1.3.1 Vectors and Vector Spaces 5
1.3.2 Complex Numbers 6

2 Sequences and Series **8**
2.1 Sequences 8
2.2 Series 10

3 Functions **11**

1 Logic, Sets and Numbers

1.1 Logic

Definition 1.1. *A **mathematical Statement** is a well-formulated assertion that is either strictly true or false.*

Definition 1.2. *A **Statement** that depends on one or multiple variables is called a **predicate**.*

Example 1.3.

$$P(x) \coloneqq x > 2$$

Definition 1.4. *The negation of a mathematical Statement A is defined as $\neg A \coloneqq$ A is false.*

Definition 1.5. *Quantifiers are short hand notation for making statements about predicates.*

$$\forall x \in \mathbb{R} : P(x)$$

$$\exists x \in \mathbb{R} : P(x)$$

$$\exists! x \in \mathbb{R} : P(x)$$

$$\nexists x \in \mathbb{R} : P(x)$$

Expresses that for all / for at least one / for exactly one or for none x in $\mathbb{R}$ satisfies the predicate.

Important 1.6. *The oderder of quantifiers matters. $\forall x \exists y$ is not equivalent to $\exists y \forall x$.*

Definition 1.7. *Basic logical connectives are defined as:*

$$A \wedge B \coloneqq A \text{ and } B \text{ are true}$$

$$A \vee B \coloneqq A \text{ or } B \text{ is true}$$

$$A \implies B \coloneqq A \text{ implies } B \text{ is true}$$

$$A \iff B \coloneqq A \text{ is equivalent to } B \text{ is true}$$

observe that equivalence holds iff. $A \implies B \wedge B \implies A$

Theorem 1.8. *The following equivalences hold:*

$$\neg(\forall x P(x)) \iff \exists x \neg P(x)$$

$$\neg(\exists x P(x)) \iff \forall x \neg P(x)$$

aswell as

$$\forall x(P(x) \wedge Q(x)) \iff (\forall x P(x)) \wedge (\forall x Q(x))$$
$$\exists x(P(x) \vee Q(x)) \iff (\exists x P(x)) \vee (\exists x Q(x))$$

1.2 Sets

Definition 1.9. *A **set** is a collection of elements, that are listed or characterized by a certain property. The empty set is denoted by $\emptyset$ and is the only set that contains no elements.*

An element x is contained in a set A iff. $x \in A$ else it is not contained in A denoted by $x \notin A$.

Definition 1.10. *Let A and B be sets. A is a subset of B denoted by $A \subseteq B$ iff. $\forall x(x \in A \implies x \in B)$.*

A is a proper subset of B denoted by $A \subset B$ iff. $A \subseteq B \wedge A \neq B$.

Definition 1.11. *Let A and B be sets.*

- *The **union** of A and B is defined as $A \cup B := \{x \mid x \in A \vee x \in B\}$.*
- *The **intersection** of A and B is defined as $A \cap B := \{x \mid x \in A \wedge x \in B\}$.*
- *The **difference** of A and B is defined as $A \setminus B := \{x \mid x \in A \wedge x \notin B\}$.*
- *The **Cartesian product** of A and B is defined as $A \times B := \{(a, b) \mid a \in A \wedge b \in B\}$.*

1.3 Numbers

The basic and often used sets of numbers are:

- $\mathbb{N}$: The set of natural numbers 1, 2, 3, 4, ...
- $\mathbb{Z}$: The set of integers ... -2, -1, 0, 1, 2, ...
- $\mathbb{Q}$: The set of rational numbers $\frac{p}{q}$ where $p, q \in \mathbb{Z}$
- $\mathbb{R}$: The set of real numbers

Remark 1.12. *Some extensions of the real numbers are* $\mathbb{R}_0^+/\mathbb{R}^+$ *and* $\mathbb{R}_0^-/\mathbb{R}^-$ *which define the intervalls* $[0,\infty)/(0,\infty)$ *and* $(-\infty,0]/(-\infty,0)$.

Definition 1.13. *Continuous subsets are called* ***intervalls.*** *An intervall is called*

- ***open*** *if it does not contain its endpoints* $I = (a,b) := \{x \mid a < x < b\}$
- ***closed*** *if it contains its endpoints* $I = [a,b] := \{x \mid a \leq x \leq b\}$
- ***half open*** *if it contains one endpoint and not the other* $I = [a,b) := \{x \mid a \leq x < b\}$

An intervall is bounded if it is bounded from above and below hence a *and* b *are finite.*

Definition 1.14. *An upper (lower) bound of a subset* $X \subset \mathbb{R}$ *is an element* $y \in \mathbb{R}$ *such that:*

$$\forall x \in X : x \leq (\geq) y$$

The maximum (minimum) of a subset $X \subset \mathbb{R}$ *is the element* $x_0 \in X$ *such that:*

$$\forall x \in X : x \leq (\geq) x_0$$

Definition 1.15. *Least upper bound and greatest lower bound are also called Supremum and Infimum.*

The ***Supremum*** $sup(X)$ *of a subset* $X \subset \mathbb{R}$ *is the least upper bound of* X*. and can be characterized by:*

$$\forall x \in X : x \leq sup(X) \wedge \forall \epsilon > 0 : \exists x \in X : x > sup(X) - \epsilon$$

The ***Infimum*** $inf(X)$ *of a subset* $X \subset \mathbb{R}$ *is the greatest lower bound of* X*. and can be characterized by:*

$$\forall x \in X : x \geq inf(X) \wedge \forall \epsilon > 0 : \exists x \in X : x < inf(X) + \epsilon$$

Example 1.16. *Let* $X = [-2,1)$ *we have a minimum of* $x_0 = -1$ *but no maximum. Observe that it is bounded from above by* 1 *and* 2 *or any number larger than* 1 *but only* 1 *is the least upper bound (Supremum).*

Theorem 1.17. *The following holds:*

- *The* ***maximum and minimum are unique*** *aslong as they exist.*

- *Every* ***non empty, bounded from above (below)*** *subset of* $\mathbb{R}$ *has a* ***maximum (minimum)****.*

Definition 1.18. *A set* X *is said to be (order)complete if for every non empty subsets* $A, B \subseteq X$

$$\forall a \in A \forall b \in B : a \leq b$$

then there there exists an element $c \in X$ *such that:*

$$\forall a \in A : a \leq c \wedge \forall b \in B : c \leq b$$

Theorem 1.19. *The Archimedian Principle says that*

- *For* $x \in \mathbb{R}$ *and* $y > 0$ *there exists an element* $n \in \mathbb{N}$ *such that* $n * y > x$*.*
- *For every* $\epsilon > 0$ *there exists an element* $n \in \mathbb{N}$ *such that* $\frac{1}{n} < \epsilon$*.*

Theorem 1.20. *The Young Inequality states that for all* $\epsilon > 0$ *and all* $x, y \in \mathbb{R}$*:*

$$2|xy| \leq \epsilon x^2 + \frac{1}{\epsilon} y^2$$

Theorem 1.21. *The real numbers* $\mathbb{R}$ *satisfy axiomatic properties for addition and multiplication (forming a field):*

- *Associativity:* $x + (y + z) = (x + y) + z$ *and* $x \cdot (y \cdot z) = (x \cdot y) \cdot z$
- *Commutativity:* $x + y = y + x$ *and* $x \cdot y = y \cdot x$
- *Neutral elements:* $\exists 0, 1 \in \mathbb{R}$
- *Inverse elements:* $-x$ *and* x^{-1} *(for* $x \neq 0$*)*
- *Distributivity:* $x \cdot (y + z) = x \cdot y + x \cdot z$

$\mathbb{R}$ *has order axioms: reflexivity, transitivity, antisymmetry, totality, making it an* ***ordered and order-complete field.***

1.3.1 Vectors and Vector Spaces

Definition 1.22. *For* $x, y \in \mathbb{R}^n$*, the standard scalar product is defined as:*

$$x \cdot y = \langle x, y \rangle \coloneqq \sum_{i=1}^{n} x_i y_i$$

The Euclidean norm is defined as:

$$||x|| := \sqrt{\sum_{i=1}^{n} x_i^2}$$

The Euclidean distance between x and y is:

$$d(x, y) := \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

Theorem 1.23. *The triangle inequality holds for all $x, y, z \in \mathbb{R}^n$:*

$$||x - z|| \leq ||x - y|| + ||y - z||$$

1.3.2 Complex Numbers

Definition 1.24. *The **complex numbers** $\mathbb{C}$ are numbers of the form:*

$$z = a + ib \quad \text{where } a, b \in \mathbb{R}$$

a is the real part $Re(z)$, and b is the imaginary part $Im(z)$. The imaginary unit i has the property $i^2 = -1$.

Definition 1.25. *Arithmetic of complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$:*

- *Addition:* $z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$
- *Multiplication:* $z_1 \cdot z_2 = (x_1x_2 - y_1y_2) + i(x_1y_2 + y_1x_2)$

The complex conjugate of $z = x + iy$ is $\bar{z} = x - iy$. Division is calculated by expanding with the conjugate of the denominator: $\frac{z}{w} = \frac{z \cdot \bar{w}}{w \cdot \bar{w}} = \frac{z \cdot \bar{w}}{|w|^2}$.

Remark 1.26. *Some usfull properties of the conjugate:*

- $z \cdot \bar{z} = |z|^2$
- $z + \bar{z} = 2Re(z)$ *and* $z - \bar{z} = 2iIm(z)$
- $\overline{z + w} = \bar{z} + \bar{w}$ *and* $\overline{z \cdot w} = \bar{z} \cdot \bar{w}$

Definition 1.27. *Geometrically, complex numbers can be represented in the Gaussian plane. The **absolute value** of $z = a + ib$ is its **distance from the origin**:*

$$|z| = \sqrt{a^2 + b^2}$$

The distance between z_1 and z_2 is $d = |z_1 - z_2|$. The triangle inequality holds in $\mathbb{C}$: $|z + w| \leq |z| + |w|$.

Theorem 1.28. *The Euler's formula states that for $t \in \mathbb{R}$:*

$$e^{it} = \cos(t) + i\sin(t)$$

This allows writing trigonometric functions using complex exponentials:

$$\cos(t) = \frac{e^{it} + e^{-it}}{2}, \quad \sin(t) = \frac{e^{it} - e^{-it}}{2i}$$

Eulers formula leads us to a new representation of complex numbers: the polar-coordinates.

Definition 1.29. *For $z \in \mathbb{C}$, the polar-coordinates are defined as:*

$$z = r(\cos(t) + i\sin(t)) = re^{it}$$

where $r = |z|$ is the absolute value and $t = \arg(z)$ is the argument.

Multiplication and division in polar-coordinates is much easier than in cartesian coordinates:

$$z_1 \cdot z_2 = r_1 e^{it_1} \cdot r_2 e^{it_2} = r_1 r_2 e^{i(t_1+t_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1 e^{it_1}}{r_2 e^{it_2}} = \frac{r_1}{r_2} e^{i(t_1-t_2)}$$

So is exponentiation or taking roots:

$$z^n = (re^{it})^n = r^n e^{int}$$

$$\sqrt[n]{z} = \sqrt[n]{r} e^{i\frac{t+2\pi k}{n}}, \quad k \in \{0, 1, \ldots, n-1\}$$

Definition 1.30. *The n-th roots of unity are the solutions to $z^n = 1$, which are given by:*

$$z_k = e^{i\frac{2\pi k}{n}}, \quad k \in \{0, 1, \ldots, n-1\}$$

Theorem 1.31. *The sum of the roots of unity is always zero.*

$$\sum_{k=0}^{n-1} e^{i\frac{2\pi k}{n}} = 0$$

Theorem 1.32. *The fundamental theorem of algebra states that every polynomial of degree $n \geq 1$ has exactly n roots in $\mathbb{C}$ (counting multiplicity). These roots appear in konplex-conjugate pairs.*

2 Sequences and Series

2.1 Sequences

Definition 2.1. *A **sequence** (Folge) $(a_n)_{n\in\mathbb{N}_0}$ is a list of numbers, mathematically a mapping $f : \mathbb{N}_0 \to X$. It can be given explicitly ($a_n = f(n)$) or recursively ($a_n = g(a_{n-1}, a_{n-2}, \dots)$).*

Definition 2.2. *A sequence (a_n) is called **convergent** with limit $L \in \mathbb{R}$ if:*

$$\forall \epsilon > 0 \exists N > 0 \text{ s.t. } \forall n > N : |a_n - L| < \epsilon$$

Written as $\lim_{n\to\infty} a_n = L$.

*If no such L exists, the sequence is **divergent**. Sequences that diverge to $\pm\infty$ are a special case of divergent sequences.*

Theorem 2.3. *The limit of a sequence is unique aslong as it exists.*

Theorem 2.4. *A subsequence $(a_{n_k})_{k\in\mathbb{N}_0}$ of a convergent sequence converges to the same limit L.*

Definition 2.5. *An **accumulation point** (Häufungspunkt) $A \in \mathbb{R}$ of a sequence fulfills:*

$$\forall \epsilon > 0 \forall N \in \mathbb{N}_0 \exists n \geq N \text{ such that } |a_n - A| < \epsilon$$

A is an accumulation point iff there exists a convergent subsequence with limit A. Every convergent sequence has exactly one accumulation point (its limit).

Theorem 2.6. *Limit arithmetic: For $\lim a_n = K$ and $\lim b_n = L$:*

- $\lim(a_n \pm b_n) = K \pm L$
- $\lim(a_n \cdot b_n) = K \cdot L$
- $\lim(a_n/b_n) = K/L$ *(if $L \neq 0$ and $b_n \neq 0$)*

Theorem 2.7. ***Sandwich-Theorem:*** *If $a_n \leq b_n \leq c_n$ for $n \geq N_0$, and $\lim a_n = \lim c_n = L$, then $\lim b_n = L$.*

Remark 2.8. *Important limits:*

- $\lim_{n\to\infty} \frac{\ln n}{n} = 0$
- $\lim_{n\to\infty} n^{1/n} = 1$
- $\lim_{n\to\infty} x^{1/n} = 1 \quad (x > 0)$
- $\lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n = e^x$
- $\lim_{n\to\infty} \frac{x^n}{n!} = 0$

Definition 2.9. *A sequence is*

- ***monotonically increasing (decreasing)*** *if $a_n \leq (\geq) a_m$ for $m > n$.*
- ***upper (lower) bounded*** *if there exists M such that $|a_n| \leq (\geq) M \quad \forall n \in \mathbb{N}_0$.*

Theorem 2.10. ***Bolzano-Weierstrass Theorem:*** *Every bounded sequence has an accumulation point.*

Theorem 2.11. ***Monotone Convergence Theorem:*** *Every bounded monotonic sequence converges to its supremum (infimum).*

Definition 2.12. *The limit of the supremum is called* ***limes superior*** *and writen as:*

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty} \sup_{k\geq n} a_k$$

The limit of the infimum is called ***limes inferior*** *and writen as:*

$$\liminf_{n\to\infty} a_n = \lim_{n\to\infty} \inf_{k\geq n} a_k$$

Definition 2.13. *A sequence is a* ***Cauchy sequence*** *if $\forall \epsilon > 0 \exists N \in \mathbb{N}$ such that $\forall m, n \geq N : |a_n - a_m| < \epsilon$. A real sequence converges iff it is a Cauchy sequence.*

Theorem 2.14. *A sequence of real numbers converges iff it is a cauchy sequence.*

2.2 Series

Definition 2.15. *A* ***series*** *(Reihe) is an infinite sum* $\sum_{i=0}^{\infty} a_i$ *of the terms of a sequence* (a_n)*.*

Theorem 2.16. ***Necessary condition for convergence:*** *for* $\sum_{i=0}^{\infty} a_i$ *to converge we need* $\lim_{n\to\infty} a_n = 0$*.*

Theorem 2.17. *For* ***series with non-negative terms*** $a_n \geq 0$*, the sequence of partial sums is monotonically increasing.* ***It converges iff it is bounded.***

Theorem 2.18. ***Comparison test*** *(Vergleichskriterium): If* $c_n \leq b_n \leq a_n$ *for* $n \geq N$*:*

- *If* $\sum a_n$ *converges, then* $\sum b_n$ *converges (**Majorantenkriterium**).*
- *If* $\sum c_n$ *diverges, then* $\sum b_n$ *diverges (**Minorantenkriterium**).*

Definition 2.19. *A series is* ***absolutely convergent*** *if* $\sum |a_n|$ *converges. Absolute convergence implies normal convergence. A* ***conditionally convergent*** *series converges, but not absolutely.*

Theorem 2.20. ***Riemann's rearrangement theorem:*** *Let* $\sum a_n$ *be a conditionally convergent series. Then for any* $x \in \mathbb{R}$*, there exists a rearrangement of the series that converges to* x*.*

Theorem 2.21. ***Leibniz Criterion****: For an alternating series* $\sum(-1)^n a_n$ *with a monotonically decreasing null sequence* (a_n)*, the series converges.*

Theorem 2.22. ***Cauchy Criterion:*** *The series* $\sum a_n$ *converges iff*

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N : |\sum_{k=n}^{m} a_k| < \epsilon$$

Theorem 2.23. *More criterions for convergence:*

- ***Root Criterion*** *(Wurzelkriterium): Let* $\rho = \limsup \sqrt[n]{|a_n|}$. *If* $\rho < 1$, *abs conv; if* $\rho > 1$, *div.*
- ***Ratio Criterion*** *(Quotientenkriterium): Let* $\rho = \lim \left|\frac{a_{n+1}}{a_n}\right|$. *If* $\rho < 1$, *abs conv; if* $\rho > 1$, *div.*

Definition 2.24. *A* ***power series*** *(Potenzreihe) centered at* a *has the form* $\sum_{k=0}^{\infty} c_k(x-a)^k$.

Theorem 2.25. *Every power series has a* ***convergence radius*** R, *with the* ***interval of convergence*** $(a-R, a+R)$. R *is computed by the formula* $R = \frac{1}{\limsup |c_n|^{1/n}}$.

Example 2.26.

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} x^n$$

$$\rho = \limsup \frac{1}{n^{1/n}} = 1 \implies R = 1$$

Therfore the series converges for

$$x \in (-1, 1)$$

.

Remark 2.27. *For* $x = 1$ *we have the harmonic series, which diverges. For* $x = -1$ *we have the alternating harmonic series, which converges.*

3 Functions

Definition 3.1. *A* ***function*** $f : D \to C$ *is a map from a domian* D *to a codomain* C *s.t. for each* $x \in D$ *there is exactly one* $y \in C$ *st.* $f(x) = y$.

Remark 3.2. *The input is called the* ***independent variable*** *(argument) while the output is called the* ***dependent variable****.*

Definition 3.3. *A function f is called*

- ***injective*** *iff* $\forall x_1, x_2 \in A : f(x_1) = f(x_2) \implies x_1 = x_2$.
- ***surjective*** *iff* $\forall y \in B \exists x \in A : f(x) = y$.
- ***bijective*** *iff f is injective and surjective.*

Example 3.4. *Let $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = x^2$. Observe that f is not surjective as no output is negative, nor is it injective as for ex. $f(1) = f(-1) = 1$.*

But $g : \mathbb{R}_0^+ \to \mathbb{R}_0^+$ with $g(x) = x^2$ is both injective and surjective.

Definition 3.5. *The* ***restriction*** *of f under $D' \subset \mathbb{D}$ is defined as $f \restriction_{D'} : D' \to C, x \mapsto f(x) \quad \forall x \in \mathbb{D}$.*

Remark 3.6. *Note that $f \restriction_{D'}$ and f are two diffrent functions.*

Definition 3.7. *Let $f : D \to C$ and $g : C \to B$ be functions then*

$$g \circ f : D \to B \quad \text{with} \quad g \circ f(x) = g(f(x)) \quad \forall x \in D$$

is called the ***composition*** *of f and g.*

Definition 3.8. *The* ***identity function*** *$id_D : D \to D$ is defined as $id_D(x) = x \quad \forall x \in D$.*

Definition 3.9. *If a function $f : D \to C$ is bijective then there exists a unique function $f^{-1} : C \to D$ such that*

$$f(f^{-1}(y)) = id_C = y \quad \forall y \in C \quad \text{and} \quad f^{-1}(f(x)) = id_D = x \quad \forall x \in D$$

f^{-1} is called the ***inverse function*** *of f.*